

ASINGO PAUL

Tech Lead | Cloud & DevOps Engineer | Software Engineer

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PROFESSIONAL SUMMARY

Tech Lead, Cloud & DevOps Engineer, and Software Engineer with hands-on experience leading the end-to-end delivery of enterprise software solutions. Experienced in translating business requirements into scalable technical architectures, leading development teams, designing cloud-native systems, automating deployments, and managing production environments.

Skilled across the full software development lifecycle, including solution architecture, backend and frontend development, cloud infrastructure, CI/CD automation, Linux administration, and production support. Passionate about building secure, scalable, and maintainable software while mentoring teams and driving successful project delivery.

TECHNICAL SKILLS

Cloud & DevOps

AWS • Docker • Kubernetes • GitHub Actions • CI/CD • Linux • Infrastructure Automation • Docker Compose

Software Engineering

Python • FastAPI • JavaScript • TypeScript • React • Next.js

Databases

PostgreSQL • SQL

Engineering

REST APIs • Microservices • System Design • Git • Agile • Performance Optimization • System Monitoring

PROFESSIONAL EXPERIENCE

Technical Lead / Lead Software Engineer

Specialized Engineering - Nairobi, Kenya

2025 – Present

Lead the technical delivery of enterprise software projects from concept to production while overseeing architecture, development, deployment, and operational support.

Key Responsibilities & Achievements

- Led the successful delivery of enterprise software solutions from requirements gathering through architecture, development, testing, deployment, and production support.
- Designed the overall system architecture to ensure scalability, maintainability, security, and long-term growth.
- Led a multidisciplinary development team, providing technical direction, coordinating delivery, conducting code reviews, and ensuring high engineering standards.
- Designed and developed backend services using FastAPI and modern frontend applications using Next.js and React.
- Built cloud-native deployment environments using Docker and Kubernetes for consistent and scalable application delivery.

- Designed and implemented CI/CD pipelines using GitHub Actions to automate testing and deployment workflows.
- Managed Linux production environments, application deployments, monitoring, troubleshooting, and ongoing maintenance.
- Worked directly with management, clients, and stakeholders to gather requirements, demonstrate progress, and align technical solutions with business objectives.
- Oversaw post-deployment support, system enhancements, and continuous improvements to maintain application reliability and performance.
- Championed modern engineering practices, including version control, containerization, automation, and collaborative development workflows.

FEATURED PROJECTS

FileHub – Enterprise Document Management System

Served as the Technical Lead responsible for the end-to-end delivery of FileHub, an enterprise document management platform designed to improve secure document storage, collaboration, and departmental workflows.

Responsibilities

- Led project planning, technical architecture, and implementation.
- Designed the complete system architecture and database structure.
- Developed core backend services and APIs.
- Built responsive frontend interfaces.
- Implemented role-based access control and permission management.
- Designed department and sub-department switching functionality.
- Integrated QR code document verification.
- Managed deployment and production environments.
- Led feature planning, implementation, testing, and production releases.
- Coordinated development activities and guided the engineering team throughout the project lifecycle.

Technologies

FastAPI • Next.js • PostgreSQL • Docker • GitHub Actions • Linux • AWS

Enterprise Cloud-Native Platform

Designed and implemented a production-ready cloud-native architecture consisting of **16 containerized microservices** supporting modular and scalable application deployment.

Highlights

- Designed the overall solution architecture.
- Built backend microservices using FastAPI.
- Developed frontend applications using Next.js.
- Containerized services using Docker.
- Automated deployments using GitHub Actions.
- Managed Linux deployment environments.
- Implemented scalable deployment workflows.

OpsBox

Designed and developed an open-source DevOps command-line utility that simplifies Linux server monitoring, operational diagnostics, and infrastructure management.

Key capabilities include:

- Server health monitoring
- CPU, memory, and disk utilization tracking
- Nginx log analysis
- JSON reporting
- Threshold-based alerts
- Automation-ready output

CERTIFICATIONS

- AWS Cloud Practitioner Essentials
- AWS AI & ML Scholars Program

EDUCATION

Bachelor of Science in Cloud Computing and Information Security

Machakos University

KEY ACHIEVEMENTS

- Led the successful delivery of enterprise software projects from design through production deployment.
- Designed and implemented scalable cloud-native architectures supporting enterprise applications.
- Built and deployed a platform comprising **16 containerized microservices**.
- Developed and released **OpsBox**, an open-source DevOps automation tool.
- Automated deployment pipelines using GitHub Actions, improving release consistency and reducing manual deployment effort.
- Successfully managed software delivery while leading cross-functional engineering teams.
- Regularly presented technical solutions and project progress to executive leadership and stakeholders.